

Python Programme List

No.	Program Name	Description
1	if Statement Program	Checks a condition and executes code only if the condition is True.
2	if-else Statement Program	Executes one block if condition is True and another block if False.
3	if-elif-else Ladder Program	Used to check multiple conditions one after another.
4	Nested if Statement Program	Uses an if statement inside another if statement.
5	match-case Statement Program	Matches a value with multiple cases similar to switch-case.
6	while Loop Program	Repeats code until the condition becomes False.
7	for Loop Program	Iterates through sequences like lists, strings, or ranges.
8	range() Function Program	Generates a sequence of numbers for looping.
9	break Statement Program	Terminates the loop immediately.
10	continue Statement Program	Skips the current iteration and continues the loop.
11	pass Statement Program	Placeholder statement that does nothing.
12	Nested Loop Program	A loop inside another loop used for patterns and matrix operations.
13	String Formatting Program	Inserts variables into strings using formatting methods.
14	String Manipulation Program	Performs operations like uppercase, lowercase, replace, split, etc.
15	List Program	Demonstrates list creation and operations in Python.
16	Tuple Program	Demonstrates tuple creation and immutable data storage.

17	Mutability of List Program	Shows that list elements can be modified after creation.
18	Immutability of Tuple Program	Shows that tuple elements cannot be changed.
19	List Comprehension Program	Creates lists using a compact and efficient syntax.
20	Right Triangle Pattern Program	Prints a right-angled triangle star pattern using loops.
21	Pyramid Pattern Program	Prints a pyramid-shaped star pattern.
22	Left Triangle Pattern Program	Prints a left-aligned triangle pattern.
23	Even Odd Number Program	Checks whether a number is even or odd.
24	Sum of Numbers Program	Calculates the sum of numbers in a range.
25	Factorial Program	Calculates factorial of a number using loops or recursion.
26	Prime Number Program	Checks whether a number is prime or not.
27	Fibonacci Series Program	Generates Fibonacci number sequence.
28	Multiplication Table Program	Prints multiplication table of a number.
29	Reverse String Program	Reverses a given string.
30	Palindrome Program	Checks whether a string or number is palindrome.
31	Armstrong Number Program	Checks whether a number is an Armstrong number.
32	Maximum Number Program	Finds the largest number among given values.
33	Minimum Number Program	Finds the smallest number among given values.
34	Sorting List Program	Sorts list elements in ascending or descending order.

35	Searching in List Program	Searches an element inside a list.
36	Count Vowels Program	Counts vowels in a string.
37	Remove Duplicate Elements Program	Removes duplicate values from a list.
38	Simple Calculator Program	Performs arithmetic operations like addition, subtraction, multiplication, division.
39	Menu Driven Program	Displays menu options and performs tasks based on user choice.
40	Student Grade Program	Calculates grades based on marks entered by user.

No.	Program Name	Description
41	Swap Two Numbers Program	Swaps values of two variables using temporary variable or without temporary variable.
42	Positive Negative Zero Program	Checks whether a number is positive, negative, or zero.
43	Largest of Three Numbers Program	Finds the largest number among three numbers.
44	Smallest of Three Numbers Program	Finds the smallest number among three numbers.
45	Leap Year Program	Checks whether a year is a leap year or not.
46	Sum of Digits Program	Calculates the sum of digits of a number.
47	Reverse Number Program	Reverses digits of a given number.
48	Decimal to Binary Program	Converts decimal number into binary format.
49	Binary to Decimal Program	Converts binary number into decimal format.
50	ASCII Value Program	Displays ASCII value of a character.

51	Character Type Checker Program	Checks whether character is alphabet, digit, or special symbol.
52	Vowel or Consonant Program	Determines whether a character is vowel or consonant.
53	Temperature Converter Program	Converts Celsius to Fahrenheit and vice versa.
54	Area of Circle Program	Calculates area of circle using radius.
55	Area of Rectangle Program	Calculates area of rectangle using length and width.
56	Area of Triangle Program	Calculates area of triangle using base and height.
57	Simple Interest Program	Calculates simple interest using formula.
58	Compound Interest Program	Calculates compound interest amount.
59	Electricity Bill Program	Calculates electricity bill based on units consumed.
60	ATM Simulation Program	Simulates ATM operations like withdraw, deposit, balance check.
61	Number Guessing Game Program	User guesses a random number generated by computer.
62	Rock Paper Scissors Game Program	Implements Rock-Paper-Scissors game logic.
63	Dice Rolling Simulator Program	Simulates random dice rolling.
64	Password Generator Program	Generates secure random passwords.
65	OTP Generator Program	Generates random OTP numbers.
66	Email Validation Program	Validates email format using conditions or regex.

67	Mobile Number Validation Program	Validates mobile number length and format.
68	Username Password Login Program	Checks username and password authentication.
69	Student Result Management Program	Calculates total, percentage, and grade of students.
70	Employee Salary Program	Calculates employee salary with bonus and deductions.
71	Bank Management Program	Performs banking operations like deposit and withdraw.
72	Shopping Bill Program	Generates bill for purchased products.
73	Restaurant Billing System Program	Calculates food order bills and taxes.
74	Library Management Program	Manages books and issue records.
75	Attendance System Program	Tracks student attendance records.
76	Voting Eligibility Program	Checks whether user is eligible for voting.
77	Age Calculator Program	Calculates age using birth year.
78	BMI Calculator Program	Calculates Body Mass Index.
79	Calendar Display Program	Displays monthly calendar using Python calendar module.
80	Stopwatch Program	Simulates stopwatch timer functionality.
81	Countdown Timer Program	Creates countdown timer using loops and time module.
82	Digital Clock Program	Displays real-time digital clock.

83	File Handling Read Program	Reads content from a file.
84	File Handling Write Program	Writes data into a file.
85	File Handling Append Program	Appends data to existing file.
86	Word Counter Program	Counts words in a text file or string.
87	Character Counter Program	Counts total characters in text.
88	Line Counter Program	Counts number of lines in a file.
89	Exception Handling Program	Handles runtime errors using try-except.
90	Custom Exception Program	Creates user-defined exceptions.
91	Dictionary Program	Demonstrates dictionary creation and operations.
92	Set Program	Demonstrates set creation and operations.
93	Set Operations Program	Performs union, intersection, difference operations.
94	Lambda Function Program	Demonstrates anonymous lambda functions.
95	Map Function Program	Applies function to all elements using map().
96	Filter Function Program	Filters elements based on condition using filter().
97	Reduce Function Program	Reduces sequence into single value.
98	Recursion Program	Demonstrates recursive function calls.
99	Class and Object Program	Demonstrates OOP concepts using classes and objects.
100	Constructor Program	Demonstrates use of constructors in classes.

101	Inheritance Program	Demonstrates inheritance between parent and child classes.
102	Polymorphism Program	Demonstrates method overloading and overriding.
103	Encapsulation Program	Demonstrates data hiding using private members.
104	Abstraction Program	Demonstrates abstract classes and methods.
105	Student CRUD Program	Performs create, read, update, delete operations for student records.
106	Contact Book Program	Stores and manages contact information.
107	To-Do List Program	Manages daily tasks and task completion.
108	Expense Tracker Program	Tracks expenses and calculates totals.
109	Quiz Application Program	Conducts multiple-choice quiz and calculates score.
110	Chatbot Program	Simulates basic chatbot conversation.